

IMSC25
SELECTED INEQUALITIES FROM LINEAR ALGEBRA

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ABSTRACT. We study selected inequalities that feature matrices in the background.

1. A MOTIVATING EXAMPLE

Problem (2015.MNG.T6). Show that the inequality

$$\frac{1}{(1-x^2)(1-y^2)(1-z^2)} + \frac{2}{(1-xy)(1-yz)(1-zx)} \geq \frac{1}{(1-x^2)(1-yz)^2} + \frac{1}{(1-y^2)(1-zx)^2} + \frac{1}{(1-z^2)(1-xy)^2}$$

holds for all real numbers $-1 < x, y, z < 1$.

2. POSITIVE SEMI-DEFINITE MATRICES

Definition. A real symmetric $n \times n$ matrix $A = (a_{ij})$ is *positive semi-definite* iff

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \geq 0$$

for all $x_1, \dots, x_n \in \mathbb{R}$. In this case, we write $A \geq 0$.

Theorem. Let A be a real symmetric $n \times n$ matrix. The following are equivalent.

- (1) A is positive semi-definite.
- (2) All eigenvalues of A are non-negative.
- (3) $A = B^T B$ for some B .
- (4) There exist vectors $u_1, \dots, u_n$ such that $a_{ij} = \langle u_i, u_j \rangle$ for all i, j .
- (5) All principal minors of A have non-negative determinant.

Example. $\begin{pmatrix} a & b \\ b & d \end{pmatrix} \geq 0$ iff $a, d \geq 0$ and $ad \geq b^2$.

Example. $\begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix} \geq 0$ iff $|a|, |b|, |c| \leq 1$ and $1 + 2abc \geq a^2 + b^2 + c^2$.

Example. $\begin{pmatrix} p & a & b \\ a & q & c \\ b & c & r \end{pmatrix} \geq 0$ then $pqr + 2abc \geq ra^2 + qb^2 + pc^2$.

Example (2015.MNG.T6). If $-1 < a_1, \dots, a_n < 1$, then

$$\left(\frac{1}{1 - a_i a_j} \right) \geq 0,$$

since $\frac{1}{1 - a_i a_j} = \langle u_i, u_j \rangle$ for $u_i = (1, a_i, a_i^2, a_i^3, \dots)$.

3. PROBLEMS

Problem 1 (2022.MNG.E6). Show that the inequality

$$\frac{3}{2-abc} \leq \frac{1}{2-a^2} + \frac{1}{2-b^2} + \frac{1}{2-c^2} \leq 1 + \frac{2}{2-abc}$$

holds for all real numbers $-1 \leq a, b, c \leq 1$ satisfying $a^2 + b^2 + c^2 = 2abc + 1$.

Problem 2 (2010.IMC.P5). Let $-1 \leq a, b, c \leq 1$ be real numbers satisfying $a^2 + b^2 + c^2 \leq 2abc + 1$. Prove the inequality

$$a^{2n} + b^{2n} + c^{2n} \leq 2(abc)^n + 1$$

for all positive integers n .

Problem 3 (2021.MNG.E6). Show that the inequality

$$a^2 + b^2 + c^2 + \frac{2ab}{1+|a-b|} \geq \frac{2bc}{1+|b+c|} + \frac{2ca}{1+|c+a|}$$

holds for all real numbers a, b and c .

Problem 4 (2021.MNG.SL). Show that the inequality

$$a^2 + b^2 + c^2 + \frac{2ab}{1+|a-b|} + \frac{2bc}{1+|b-c|} + \frac{2ca}{1+|c-a|} \geq \left(\frac{a}{1+|b+c|} + \frac{b}{1+|c+a|} + \frac{c}{1+|a+b|} \right)^2$$

holds for all real numbers a, b and c .

Problem 5 (2021.MNG.SL). Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n \frac{a_i a_j}{1+|a_i - a_j|} \geq 0$$

holds for all real numbers $a_1, \dots, a_n$.

Problem 6 (2021.IMO.P2). Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i - x_j|} \leq \sum_{i=1}^n \sum_{j=1}^n \sqrt{|x_i + x_j|}$$

holds for all real numbers $x_1, \dots, x_n$.

4. CONDITIONALLY NEGATIVE DEFINITE MATRICES

Definition. A real symmetric $n \times n$ matrix $A = (a_{ij})$ is *conditionally negative definite* iff

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \leq 0$$

for all $x_1, \dots, x_n \in \mathbb{R}$ satisfying $x_1 + \dots + x_n = 0$.

Theorem. Let $A = (a_{ij})$ be a real symmetric $n \times n$ matrix with zeros on the diagonal. The following are equivalent.

- (1) A is conditionally negative definite.
- (2) There exist vectors $u_1, \dots, u_n$ such that $a_{ij} = \|u_i - u_j\|^2$ for all i, j .

Problem 7 (2023.MNG.T6). Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n x_i x_j \sqrt{1 - x_i^2 x_j^2} \leq 0$$

holds for all real numbers $-1 < x_1, \dots, x_n < 1$ satisfying $x_1 + \dots + x_n = 0$.

Problem 8 (2023.MNG.SL). Show that the inequality

$$\sum_{i=1}^n \sum_{j=1}^n x_i x_j \sqrt{|x_i - x_j|} \leq 0$$

holds for all real numbers $x_1, x_2, \dots, x_n$ satisfying $x_1 + \dots + x_n = 0$.